

Data Structures Implementation (C++) :

Objective :

The objective of this project is to implement core data structures from scratch in C++ without using STL containers. The project demonstrates insertion, deletion, and search operations for Stack and Queue data structures.

Data Structures Implemented :

1. Stack

A Stack is a linear data structure that follows the Last In First Out (LIFO) principle.

Operations implemented:

- Insert (Push)
- Delete (Pop)
- Search
- Display

2. Queue

A Queue is a linear data structure that follows the First In First Out (FIFO) principle.

Operations implemented:

- Insert (Enqueue)
- Delete (Dequeue)
- Search
- Display

Development Environment :

- Programming Language: C++
- IDE: Visual Studio Code
- Compiler: G++
- Operating System: Windows

Compile and Run Commands :

Stack

Compile:

```
g++ Stack.cpp -o Stack
```

Run:

.\Stack.exe

Queue

Compile:

g++ Queue.cpp -o Queue

Run:

.\Queue.exe

Sample Input and Output:

Stack

Operations Performed:

1. Insert 10
2. Insert 20
3. Insert 30
4. Search 20
5. Delete Top Element
6. Display Stack

Output:

10 inserted into stack.
20 inserted into stack.
30 inserted into stack.
Stack elements: 30 20 10
20 found in stack.
30 deleted from stack.
Stack elements: 20 10

Queue

Operations Performed:

1. Insert 10

2. Insert 20
3. Insert 30
4. Search 20
5. Delete Front Element
6. Display Queue

Output:

10 inserted into queue.
20 inserted into queue.
30 inserted into queue.
Queue elements: 10 20 30
20 found in queue.
10 deleted from queue.
Queue elements: 20 30

Time Complexity Analysis :

Stack

<u>Operation</u>	<u>Complexity</u>
Push	$O(1)$
Pop	$O(1)$
Search	$O(n)$

Queue

<u>Operation</u>	<u>Complexity</u>
Enqueue	$O(1)$
Dequeue	$O(1)$
Search	$O(n)$

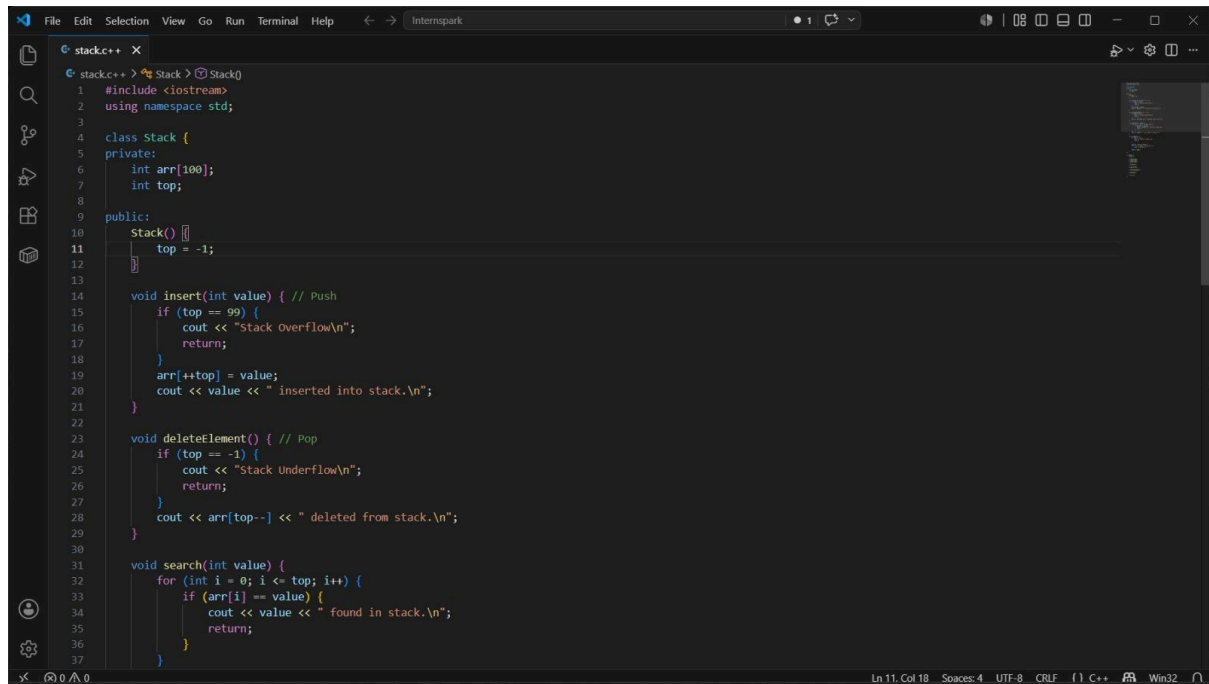
Conclusion :

This project successfully implements Stack and Queue data structures from scratch in C++. All required operations including insertion, deletion, and searching were implemented manually without using STL containers. The driver programs demonstrate the functionality of each data structure, and complexity analysis was performed to evaluate their efficiency.

Screenshots:

Insert the following screenshots here:

Stack.cpp source code



```
1 #include <iostream>
2 using namespace std;
3
4 class Stack {
5 private:
6     int arr[100];
7     int top;
8
9 public:
10    Stack() {
11        top = -1;
12    }
13
14    void insert(int value) { // Push
15        if (top == 99) {
16            cout << "Stack Overflow\n";
17            return;
18        }
19        arr[++top] = value;
20        cout << value << " inserted into stack.\n";
21    }
22
23    void deleteElement() { // Pop
24        if (top == -1) {
25            cout << "Stack Underflow\n";
26            return;
27        }
28        cout << arr[top--] << " deleted from stack.\n";
29    }
30
31    void search(int value) {
32        for (int i = 0; i <= top; i++) {
33            if (arr[i] == value) {
34                cout << value << " found in stack.\n";
35                return;
36            }
37        }
38    }
39 }
```

```
File Edit Selection View Go Run Terminal Help
stack.cpp X
stack.cpp Stack > deleteElement()
4 class Stack {
23 void deleteElement() { // Pop
24     top--;
25     cout << arr[top] << " deleted from stack.\n";
26 }
27
28 void search(int value) {
29     for (int i = 0; i <= top; i++) {
30         if (arr[i] == value) {
31             cout << value << " found in stack.\n";
32             return;
33         }
34     }
35     cout << value << " not found in stack.\n";
36 }
37
38 void display() {
39     if (top == -1) {
40         cout << "Stack is empty.\n";
41         return;
42     }
43     cout << "Stack elements: ";
44     for (int i = top; i >= 0; i--) {
45         cout << arr[i] << " ";
46     }
47     cout << endl;
48 }
49 };
50
51 int main() {
52     Stack s;
53
54     s.insert(10);
55     s.insert(20);
56     s.insert(30);
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58     s.display();
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```

```
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Queue.c++ X
Queue.c++ > Queue > deleteElement()
1 #include <iostream>
2 using namespace std;
3
4 class Queue {
5 private:
6     int arr[100];
7     int front, rear;
8
9 public:
10    Queue() {
11        front = 0;
12        rear = -1;
13    }
14
15    void insert(int value) { // Enqueue
16        if (rear == 99) {
17            cout << "Queue Overflow\n";
18            return;
19        }
20
21        arr[++rear] = value;
22        cout << value << " inserted into queue.\n";
23    }
24
25    void deleteElement() { // Dequeue
26        if (front > rear) {
27            cout << "Queue Underflow\n";
28            return;
29        }
30
31        cout << arr[front++] << " deleted from queue.\n";
32    }
33
34    void search(int value) {
35        for (int i = front; i <= rear; i++) {
36            if (arr[i] == value) {
37                cout << value << " found in queue.\n";
38            }
39        }
40    }
41
42    void display() {
43        if (front > rear) {
44            cout << "Queue is empty.\n";
45            return;
46        }
47
48        cout << "Queue elements: ";
49        for (int i = front; i <= rear; i++) {
50            cout << arr[i] << " ";
51        }
52        cout << endl;
53    }
54
55    ~Queue() {}
56
57 };
58
59 int main() {
60     Queue q;
61 }
```

```
File Edit Selection View Go Run Terminal Help Interspark
Queue.c++ X
Queue.c++ > Queue > deleteElement()
4 class Queue {
25 void deleteElement() { // Dequeue
26     if (front > rear) {
27         cout << "Queue Underflow\n";
28         return;
29     }
30
31     cout << arr[front++] << " deleted from queue.\n";
32 }
33
34 void search(int value) {
35     for (int i = front; i <= rear; i++) {
36         if (arr[i] == value) {
37             cout << value << " found in queue.\n";
38             return;
39         }
40     }
41
42     cout << value << " not found in queue.\n";
43 }
44
45 void display() {
46     if (front > rear) {
47         cout << "Queue is empty.\n";
48         return;
49     }
50
51     cout << "Queue elements: ";
52     for (int i = front; i <= rear; i++) {
53         cout << arr[i] << " ";
54     }
55     cout << endl;
56 }
57 };
58
59 int main() {
60     Queue q;
61 }
```

The screenshot shows a C++ IDE with a file named `Queue.cpp` open. The code defines a `Queue` class with methods `insert`, `display`, `search`, and `deleteElement`. The `main` function demonstrates the usage of the queue by inserting 10, 20, and 30, displaying the queue, searching for 20, deleting an element, and displaying the queue again.

```
1  Queue.cpp > Queue > deleteElement()
2
3  class Queue {
4      void display() {
5          // ...
6      }
7  };
8
9  int main() {
10     Queue q;
11
12     q.insert(10);
13     q.insert(20);
14     q.insert(30);
15
16     q.display();
17     q.search(20);
18     q.deleteElement();
19     q.display();
20
21     return 0;
22 }
```

Stack program execution/output

The screenshot shows a C++ IDE with a file named `Stack.cpp` open. The code defines a `Stack` class with methods `insert` and `display`. The `main` function demonstrates the usage of the stack by inserting 10, 20, and 30, displaying the stack, finding an element, deleting an element, and displaying the stack again.

```
1  Stack.cpp > main()
2  #include <iostream>
3  using namespace std;
4
5  class Stack {
6  private:
7      int arr[100];
8      int top;
9  public:
10     Stack() {
11         top = -1;
12     }
13
14     void insert(int value) { // Push
15         if (top == 99) {
16             cout << "Stack overflow\n";
17         }
18         arr[++top] = value;
19     }
20
21     void display() {
22         for (int i = 0; i <= top; i++)
23             cout << arr[i] << " ";
24         cout << endl;
25     }
26
27     int find(int value) {
28         for (int i = 0; i <= top; i++)
29             if (arr[i] == value)
30                 return i;
31         return -1;
32     }
33
34     void deleteElement() {
35         arr[top--] = 0;
36     }
37 }
```

The terminal output shows the execution of the program:

```
PS C:\Users\Asus\OneDrive\Desktop\Internspark> cd "C:\Users\Asus\OneDrive\Desktop\Internspark\" ; if ($?) { g++ stack.cpp -o stack } ; if ($?) { .\stack }
10 inserted into stack.
20 inserted into stack.
30 inserted into stack.
Stack elements: 30 20 10
20 found in stack.
30 deleted from stack.
Stack elements: 20 10
PS C:\Users\Asus\OneDrive\Desktop\Internspark>
```

Queue program execution/output

The image shows a Visual Studio Code editor window with a C++ file named `Queue.cpp` and a terminal window below it.

Queue.cpp

```
1 #include <iostream>
2 using namespace std;
3
4 class Queue {
5 private:
6     int arr[100];
7     int front, rear;
8
9 public:
10    Queue() {
11        front = 0;
12        rear = -1;
13    }
14
15    void insert(int value) { // Enqueue
16        if (rear == 99) {
```

Terminal

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Asus\OneDrive\Desktop\Internspark> cd "C:\Users\Asus\OneDrive\Desktop\Internspark\" ; if ($?) { g++ Queue.cpp -o Queue } ; if ($?) { .\Queue }
10 inserted into queue.
20 inserted into queue.
30 inserted into queue.
Queue elements: 10 20 30
20 found in queue.
10 deleted from queue.
Queue elements: 20 30
PS C:\Users\Asus\OneDrive\Desktop\Internspark>
```

The terminal output shows the execution of the `Queue` program. It inserts 10, 20, and 30 into the queue, then prints the elements (10 20 30). It then finds 20 in the queue and deletes 10 from the queue, resulting in the queue elements being 20 30.